Incomplete quadratic equations

When b or c is zero the formula is unnecessary — three short methods that are quicker and safer.

LESSON 64–65 | 2 × 40 MIN | ALGEBRA 8, CHAPTER III §21 | STAGE 9 · BEYOND

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Recognise the three incomplete forms.
- Solve $ax^2 + c = 0$ by taking roots.
- Solve $ax^2 + bx = 0$ by taking out a common factor.
- Explain why you must never divide an equation by x.

TEXTBOOK

National	Chapter III, pp. 129–133
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

The three forms

FORM	MISSING	METHOD	ROOTS
$ax^2 = 0$	b and c	divide by a	$x = 0$ only
$ax^2 + bx = 0$	c	common factor x	$x = 0$ and $x = -\frac{b}{a}$
$ax^2 + c = 0$	b	make x^2 the subject	two, or none

Form 1 — no b

$$ax^2 + c = 0 \Longrightarrow x^2 = -\frac{c}{a}$$

- If $-\frac{c}{a} > 0$ there are **two roots**, $x = \pm\sqrt{-c/a}$.
- If it is 0, one root, $x = 0$.
- If it is **negative**, there are **no roots** — no real number squares to a negative.

$$2x^2 - 18 = 0 \Longrightarrow x^2 = 9 \Longrightarrow x = \pm 3$$

$$x^2 + 4 = 0 \Longrightarrow x^2 = -4 \Longrightarrow no\ roots$$

DO NOT LOSE THE MINUS ROOT

$x^2 = 9$ has **two** answers, 3 and -3. The symbol $\sqrt{9}$ means only 3 — that is why the ± has to be written by hand.

Form 2 — no c

$$ax^2 + bx = 0 \Longrightarrow x(ax + b) = 0 \Longrightarrow x = 0 \text{ or } x = -\frac{b}{a}$$

$$3x^2 - 12x = 0 \Longrightarrow 3x(x - 4) = 0 \Longrightarrow x = 0 \text{ or } x = 4$$

NEVER DIVIDE BY X

Dividing $3x^2 = 12x$ by x gives $3x = 12$, so $x = 4$ — and the root $x = 0$ has vanished. You may only divide by something you know is not zero, and x might be. Always move everything to one side and factorise instead.

An equation of this form **always** has $x = 0$ as a root, because the free term is zero.

02 Worked examples

EXAMPLE 1 ***Solve $5x^2 - 45 = 0$***

1. $5x^2 = 45$
2. $x^2 = 9$

   Divide by 5.
3. $x = 3$ or $x = -3$

   Both square roots.

ANSWER

$x = \pm 3$

EXAMPLE 2 ***Solve $4x^2 + 7x = 0$***

1. $x(4x + 7) = 0$

   Common factor x.
2. $x = 0$ or $4x + 7 = 0$

   Zero product rule.
3. $x = 0$ or $x = -1.75$

ANSWER

$x = 0, x = -1.75$

EXAMPLE 3 ***Solve $3x^2 + 12 = 0$***

1. $3x^2 = -12$
2. $x^2 = -4$
3. A square is never negative.
4. No real roots.

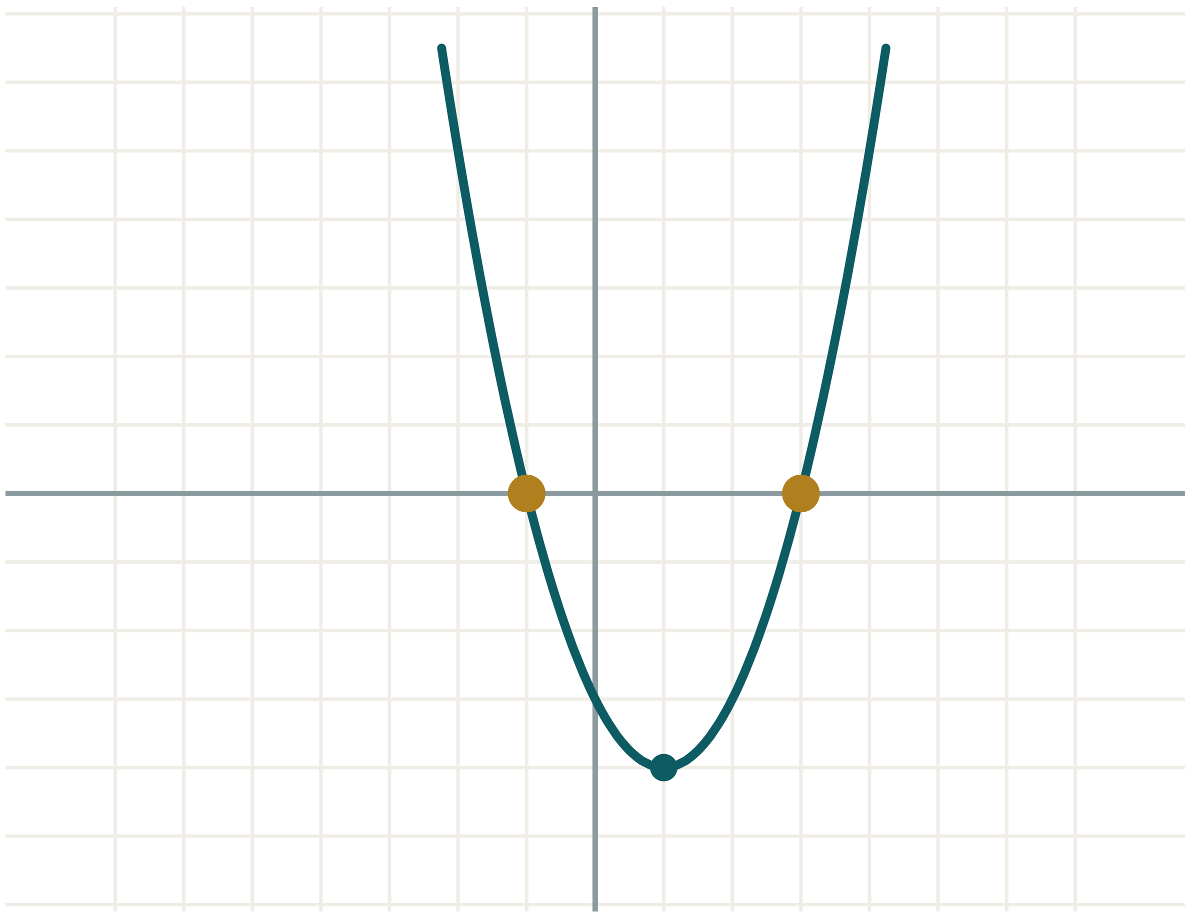
ANSWER

No roots.

03 Interactive model

Set b = 0 and vary c, then set c = 0 and vary b — the two incomplete shapes appear on the graph.

Set b or c to zero



a **1** | b **-2** | c **-3** | $D = b^2 - 4ac$ **16** | roots **two** | x_1 **-1** | x_2 **3** | $x_1 + x_2$ **2** | $x_1 \cdot x_2$ **-3**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Incomplete quadratic equation	Chala kvadrat tenglama	Неполное квадратное уравнение
Common factor	Umumiy ko‘paytuvchi	Общий множитель
Take the square root	Kvadrat ildiz chiqarish	Извлечь квадратный корень
Two opposite roots	Qarama-qarshi ildizlar	Противоположные корни
Lose a root	Ildizni yo‘qotish	Потерять корень
No real roots	Haqiqiy ildizga ega emas	Нет действительных корней
Zero root	Nol ildiz	Нулевой корень

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$x^2 = 16$ gives:

- $x = 4$
- $x = -4$
- $x = \pm 4$
- $x = 8$

$x^2 + 9 = 0$ has:

- $x = \pm 3$
- $x = 3$
- no roots
- $x = 0$

$x^2 - 6x = 0$ gives:

- $x = 6$
- $x = 0, x = 6$
- $x = \pm 6$
- $x = 0$

Dividing $x^2 = 5x$ by x is wrong because:

- it is too slow
- it loses the root $x = 0$
- x^2 is not divisible by x
- the answer becomes negative

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Solve $x^2 = 25$

   ANSWER $x = \pm 5$

2. Solve $x^2 - 49 = 0$

   ANSWER $x = \pm 7$

3. Solve $x^2 + 1 = 0$

   ANSWER No roots.

4. Solve $x^2 = 0$

   ANSWER $x = 0$

5. Solve $x^2 - 3x = 0$

   ANSWER $x = 0, x = 3$

6. Solve $x^2 + 5x = 0$

   ANSWER $x = 0, x = -5$

7. Solve $2x^2 = 8$

   ANSWER $x = \pm 2$

Medium · the standard question

7 PROBLEMS

1. Solve $5x^2 - 45 = 0$

   ANSWER $x = \pm 3$

2. Solve $4x^2 + 7x = 0$

   ANSWER $x = 0, x = -1.75$

3. Solve $3x^2 + 12 = 0$

   ANSWER No roots.

4. Solve $3x^2 - 12x = 0$

   ANSWER $x = 0, x = 4$

5. Solve $9x^2 - 4 = 0$

   ANSWER $x = \pm\frac{2}{3}$

6. Solve $x^2 = 7x$

   ANSWER $x = 0, x = 7$

7. Solve $2x^2 - 50 = 0$

   ANSWER $x = \pm 5$

Hard · stretch and reason

7 PROBLEMS

1. *Solve* $(x-2)^2 = 9$

   ANSWER $x - 2 = \pm 3$: $x = 5$ or $x = -1$

2. *Solve* $4x^2 = 3x$

   ANSWER $x(4x - 3) = 0$: $x = 0$ or $x = 0.75$

3. *For which* c *does* $x^2 + c = 0$ *have two roots?*

   ANSWER $c < 0$

4. *Solve* $2x^2 - 6 = 0$ *and give the answer exactly.*

   ANSWER $x^2 = 3$, so $x = \pm\sqrt{3}$

5. *Solve* $x(x + 3) = 4x$

   ANSWER $x^2 + 3x - 4x = 0 \Longrightarrow x(x - 1) = 0$: $x = 0, x = 1$

6. *The area of a square is 121 cm². Find its side, and say why only one answer makes sense.*

   ANSWER $x^2 = 121$ gives $x = \pm 11$; a length cannot be negative, so $11\ cm$.

7. *Solve* $(2x - 1)^2 = 0$

   ANSWER $2x - 1 = 0$, so $x = 0.5$ — one root only.

07 Homework

Homework — 6 problems

Algebra 8, Chapter III, pp. 129–133. Never divide by x.

1. *Solve* $x^2 - 64 = 0$
2. *Solve* $3x^2 - 27 = 0$
3. *Solve* $x^2 + 16 = 0$
4. *Solve* $5x^2 + 20x = 0$
5. *Solve* $x^2 = 9x$

6. *Solve* $(x + 1)^2 = 16$